Indefinite Integration

Introduction

If g'(x) = f (x) →
then g (x) is called the
antiderivative of f (x)

$$\frac{d}{dx}x^3 \xrightarrow{\text{Differentiation}} 3x^2$$

$$3x^2 \xrightarrow{\text{Integration}} x^3$$

$$\sin x \xrightarrow{\text{Differentiation}} \cos x$$

$$\cos x \xrightarrow{\text{Integration}} \sin x$$

$$\int 3x^2\,dx = x^3 + c$$

Constant of Integration ← **Very important to mention**

Fundamental Formulas

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{1}{x} \, dx = \log_e |x| + C$$

$$\int a^x \, dx = \frac{a^x}{\log_e a} + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \sec x \, \tan x \, dx = \sec x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \csc x \, \cot x \, dx = -\csc x + C$$

$$\int \sinh(x)\,dx = \cosh(x) + C$$

$$\int \cosh(x)\,dx = \sinh(x) + C$$

$$\int \operatorname{sech}^2(x)\,dx = \tanh(x) + C$$

$$\int \operatorname{cosech}^2(x)\,dx = -\coth(x) + C$$

$$\int \operatorname{sech}(x)\tanh(x)\,dx = -\operatorname{sech}(x) + C$$

$$\int \operatorname{cosech}(x)\coth(x)\,dx = -\operatorname{cosech}(x) + C$$

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1} x + C$$

$$\int \frac{-1}{\sqrt{1-x^2}}\,dx = \cos^{-1} x + C$$

$$\int \frac{1}{1+x^2}\,dx = \tan^{-1} x + C$$

$$\int \frac{1}{|x|\sqrt{x^2-1}}\,dx = \sec^{-1} x + C$$

$$\int \frac{-1}{|x|\sqrt{x^2-1}}\,dx = \operatorname{cosec}^{-1} x + C$$

Constant Rule (k = Constant)

$$\int k.f(x).dx = k\int f(x).dx$$

Example →

$$\int 2\sin x.dx = 2\int \sin x.dx = -2\cos x$$

Distribution

$$\int [f(x) \pm g(x)]dx = \int f(x).dx \pm \int g(x).dx$$

Integration by Substitution

$$I = \int f(g(x)).g'(x).dx$$

$$\text{Let } g(x) = t$$

$$\text{Differentiate } g(x) = t \text{ w.r.t } x \rightarrow$$

$$\frac{d}{dx}g(x) = g'(x) = \frac{dt}{dx} \rightarrow dt = g'(x).dx$$

$$I=\int f\left(g\left(x\right)\right).\,g'\left(x\right).\,dx=\int f\left(t\right).\,dt$$

Example →

$$I=\int \frac{\left(\log_e x\right)^2}{x}dx$$

$$Let\ \log_e x=t$$

$$\frac{d}{dx}\left(\log_e x\right)=\frac{dt}{dx}\rightarrow\frac{1}{x}=\frac{dt}{dx}$$

$$\frac{dx}{x}=dt$$

$$I=\int \frac{\left(\log_e x\right)^2}{x}dx=\int \left(\log_e x\right)^2\frac{dx}{x}$$

$$I=\int \left(\log_e x\right)^2\frac{dx}{x}=\int t^2.\,dt$$

$$\int t^2.\,dt=\frac{t^3}{3}+c=\frac{\left(\log_e x\right)^3}{3}+c$$

Fundamental Formulas for Integration

$$\boxed{\int \tan x . dx = \log_e |\sec x| + c}$$

$$\int \tan x . dx = \int \frac{\sin x}{\cos x} dx$$

$$Let \ \cos x = t \rightarrow -\sin x . dx = dt$$

$$\int \frac{\sin x}{\cos x} dx = \int \frac{-dt}{t} = -\log_e t + c$$

$$-\log_e t + c = -\log_e |\cos x| + c$$

$$= \log_e |\cos x|^{-1} + c = \log_e |\sec x| + c$$

$$\boxed{\begin{gathered} \int \cot x \, dx = -\log_e |\mathrm{cosec}\, x| + c \\ \int \cot x \, dx = \log_e |\sin x| + c \end{gathered}}$$

$$\boxed{\begin{aligned}\int \sec x\,dx &= \log_e|\sec x + \tan x| + c\\ &= \log_e\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right| + c\end{aligned}}$$

$$\int \sec x\,dx = \int \frac{\sec x(\sec x + \tan x)}{(\sec x + \tan x)}\,dx$$

$$(\sec x + \tan x) = t$$

$$(\sec x \tan x + \sec^2 x)\,dx = dt$$

$$\int \sec x\,dx = \int \frac{\sec^2 x + \sec x \tan x}{(\sec x + \tan x)}\,dx$$

$$= \int \frac{dt}{t} = \log_e t + c = \log_e(\sec x + \tan x) + c$$

$$\implies \sec x + \tan x = \frac{1+\sin x}{\cos x}$$

$$= \frac{\cos^2\frac{x}{2} + \sin^2\frac{x}{2} + 2\sin\frac{x}{2}\cos\frac{x}{2}}{\cos^2\frac{x}{2} - \sin^2\frac{x}{2}}$$

$$= \frac{\left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)^2}{\left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)\left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)}$$

$$= \frac{\left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)}{\left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)}$$

$$= \frac{\frac{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)}{\cos\frac{x}{2}}}{\frac{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)}{\cos\frac{x}{2}}} = \frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}$$

$$\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}} = \frac{\tan\frac{\pi}{4}+\tan\frac{x}{2}}{1-\tan\frac{\pi}{4}\tan\frac{x}{2}}$$

$$\frac{\tan\frac{\pi}{4}+\tan\frac{x}{2}}{1-\tan\frac{\pi}{4}\tan\frac{x}{2}} = \tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$$

$$\implies \sec x + \tan x = \tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$$

Hence Proved.

$$\int \text{cosec } x \cdot dx = \log_e |\csc x - \cot x| + c$$

$$= -\log_e |\csc x + \cot x| + c$$

$$= \log_e \left|\tan\left(\frac{x}{2}\right)\right| + c$$

Integrals of the form →

$$\int f(ax+b)\,dx$$

In the case of such integrals, just divide the answer obtained by *a* (coefficient of x) →

For example:

$$\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c$$

$$\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + c$$

$$\int \frac{1}{ax+b}\,dx = \frac{\log_e|ax+b|}{a} + c$$

$$\int a^{bx+c}\,dx = \frac{a^{bx+c}}{b\log_e a} + c$$

$$\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + c$$

$$\int \sin(ax+b)\,dx = -\cos(ax+b) + c$$

$$\int \sec^2(ax+b)\,dx = \frac{\tan(ax+b)}{a} + c$$

$$\int \operatorname{cosec}^2(ax+b)\,dx = \frac{-\cot(ax+b)}{a} + c$$

$$\int \sec(ax+b)\ \tan(ax+b)\,dx$$

$$= \frac{\sec(ax+b)}{a} + c$$

$$\int \csc(ax+b)\ \cot(ax+b)\,dx$$

$$= \frac{-\csc(ax+b)}{a} + c$$

Standard Substitutions to Make Problem Solving Easier

$$a^2 - x^2 \implies x = a\sin(\theta) \text{ or } a\cos(\theta)$$

$$a^2 - x^2 \implies x = a\sin(\theta) \text{ or } a\cos(\theta)$$

$$x^2 - a^2 \implies x = a\sec(\theta) \text{ or } a\csc(\theta)$$

$$\sqrt{\frac{a+x}{a-x}}, \sqrt{\frac{a-x}{a+x}} \Rightarrow x = a\cos\left(2\theta\right)$$

Integrals of Some Special Functions

$$\boxed{\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + c}$$

Proof→

$$x = a\sin\left(\theta\right)$$

$$dx = a\cos\left(\theta\right)d\theta$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \int \frac{a\cos\theta \, d\theta}{\sqrt{a^2 - a^2\sin^2\theta}}$$

$$= \int \frac{a\cos\theta\, d\theta}{a\cos\theta}$$

$$= \int d\theta = \theta + c = \sin^{-1}\frac{x}{a} + c$$

Hence Proved.

$$\boxed{\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c}$$

$$x = a\tan(\theta)$$

$$dx = a\sec^2(\theta)\, d\theta$$

$$\int \frac{dx}{a^2 + x^2} = \int \frac{a\sec^2(\theta)\, d\theta}{a^2 + a^2\tan^2\theta}$$

$$= \int \frac{a\sec^2(\theta)\, d\theta}{a^2\sec^2(\theta)} = \frac{1}{a}\theta + c$$

$$= \frac{1}{a}\tan^{-1}\frac{x}{a} + c$$

Hence Proved.

Integration for Some Special Functions

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \log_e \left| x + \sqrt{a^2+x^2} \right| + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \log_e \left| x + \sqrt{x^2-a^2} \right| + C$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \log_e \left| \frac{a+x}{a-x} \right| + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log_e \left| \frac{x-a}{x+a} \right| + C$$

Integration by Parts (ILATE Rule)

$$\boxed{\begin{aligned} \int u.\,v.\,dx = u \int v.\,dx \\ - \int \left[\frac{du}{dx} \left(\int v.\,dx \right) \right] dx \end{aligned}}$$

The ILATE rule helps you choose which function becomes u.
You always choose u based on the following priority order:

1. **I → Inverse Trigonometric Functions**
   - **E.g. → $\sin^{-1}x$**
2. **L → Logarithmic Functions**
   - **E.g. → $\log_e|x| \to \ln|x|$**
3. **A → Algebraic Functions**
   - **E.g. → $x^2 + 1$**
4. **T → Trigonometric Functions**
   - **E.g. → $\sin x$**
5. **E → Exponential Functions**
   - **E.g. → e^x**

$$I = \int \log_e x = \int \underset{\substack{\downarrow\\ \textbf{3. A}\\ \downarrow\\ v}}{(1)}\ \underset{\substack{\downarrow\\ \textbf{2. L}\\ \downarrow\\ u}}{\log_e x}$$

$$\int (1)\log_e x = \log_e x \int 1\,dx - \int \left[\frac{d}{dx}\log_e x \left(\int 1\,dx\right)\right] dx$$

Integrals of the Following Forms

$$\boxed{\int e^{ax}\sin bx\,dx \text{ or } \int e^{ax}\cos bx\,dx}$$

$$I = \int e^{2x}\cos x\,dx$$

5. E

v

4. T

u

$$\int e^{2x}\cos x\,dx = \cos x \int e^{2x}\,dx - \int \left[\frac{d}{dx}\cos x \left(\int e^{2x}\,dx\right)\right] dx$$

$$\int e^{2x}\cos x\,dx = \cos x\,\frac{e^{2x}}{2} - \int -\sin x\,\frac{e^{2x}}{2}\,dx$$

$$\int e^{2x}\cos x\,dx = \frac{1}{2}\cos x\,e^{2x} + \frac{1}{2}\int \underset{\substack{\downarrow\\ \text{4. T}\\ \downarrow\\ u}}{\sin x}\; \underset{\substack{\downarrow\\ \text{5. E}\\ \downarrow\\ v}}{e^{2x}}\,dx$$

$$\int \sin x\,e^{2x}\,dx = \sin x\,\frac{e^{2x}}{2} - \int\left[\frac{d}{dx}\sin x\left(\int e^{2x}\right)\right]dx$$

$$\int \sin x\, e^{2x}\, dx = \frac{1}{2}\sin x\, e^{2x} - \frac{1}{2}\int \cos x\, e^{2x}\, dx$$

$$\int \sin x\, e^{2x}\, dx = \frac{1}{2}\sin x\, e^{2x} - \frac{1}{2} I$$

$$I = \int e^{2x} \cos x\, dx = \frac{1}{2}\cos x\, e^{2x} + \frac{1}{2}\left(\frac{1}{2}\sin x\, e^{2x} - \frac{1}{2} I\right)$$

$$\frac{5}{4} I = \frac{5}{4}\int e^{2x} \cos x\, dx = \frac{1}{2}\cos x\, e^{2x} + \frac{1}{4}\sin x\, e^{2x}$$

$$I = \int e^{2x} \cos x\, dx = \frac{2}{5}\cos x\, e^{2x} + \frac{1}{5}\sin x\, e^{2x} + c$$

Some Special Integrals

$$\int e^{x}\left[f(x)+f'(x)\right]dx = e^{x}f(x)+c$$

This can be proved by ILATE rule.

$$\int \left[f(x)+x\,f'(x)\right]dx = x\,f(x)+c$$

Some More Special Integrals

$$\int \sqrt{x^{2}+a^{2}}\,dx = \frac{1}{2}x\sqrt{x^{2}+a^{2}} + \frac{1}{2}a^{2}\ln\left|x+\sqrt{x^{2}+a^{2}}\right| + C$$

Proof → Put x = a tan θ

$$\int \sqrt{x^2 + a^2}\, dx = \frac{1}{2} x \sqrt{x^2 + a^2} + \frac{a^2}{2} \sinh^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \sqrt{x^2 - a^2}\, dx = \frac{1}{2} x \sqrt{x^2 - a^2} - \frac{1}{2} a^2 \ln\left|x + \sqrt{x^2 - a^2}\right| + C$$

Proof → Put x = a cot θ

$$\int \sqrt{a^2 - x^2}\, dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{1}{2} a^2 \sin^{-1}\left(\frac{x}{a}\right) + C$$

Proof → Put x = a sin θ

$$\int \frac{1}{\sqrt{x^2+a^2}}dx = \sinh^{-1}\left(\frac{x}{a}\right)+C$$

$$\int \frac{1}{\sqrt{a^2-x^2}}dx = \sin^{-1}\left(\frac{x}{a}\right)+C$$

$$\int \frac{1}{\sqrt{x^2-a^2}}dx = \cosh^{-1}\left(\frac{x}{a}\right)+C$$

$$\int \frac{1}{a^2+x^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right)+C$$

$$\int \frac{dx}{x^2-a^2} = -\frac{1}{a}\coth^{-1}\left(\frac{x}{a}\right)+C$$

$$\sinh^{-1}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$$

$$\cosh^{-1}(x) = \ln\left(x + \sqrt{x^2 - 1}\right)$$

$$\tanh^{-1}(x) = \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right)$$

$$\coth^{-1}(x) = \frac{1}{2}\ln\left(\frac{x + 1}{x - 1}\right)$$

$$\sinh^{-1}\left(\frac{x}{a}\right) = \ln\left(x + \sqrt{x^2 + a^2}\right) - \ln(a)$$

$$\cosh^{-1}\left(\frac{x}{a}\right) = \ln\left(x + \sqrt{x^2 - a^2}\right) - \ln(a)$$

$$\tanh^{-1}\left(\frac{x}{a}\right) = \frac{1}{2}\ln\left(\frac{a + x}{a - x}\right)$$

$$\coth^{-1}\left(\frac{x}{a}\right) = \frac{1}{2}\ln\left(\frac{x + a}{x - a}\right)$$

Different Forms of Special Integrals

FORM – 1

$$\int \frac{1}{ax^2 + bx + c}\, dx$$

$$\int \frac{1}{\sqrt{ax^2 + bx + c}}\, dx$$

$$\int \sqrt{ax^2 + bx + c}\, dx$$

In problems like these, where there is a 2^{nd} degree polynomial → Use Completing the Square Method.

EXAMPLE →

$$\int \sqrt{x^2 + 2x - 3}\,.\,dx$$

$$= \int \sqrt{(x+1)^2 - 4}\,.\,dx$$

$$\left(\text{put } t = x + 1 \quad dt = dx\right)$$

$$= \int \sqrt{t^2 - 4}\,.\,dt$$

$$= \frac{1}{2} t\sqrt{t^2 - 4}$$

$$-2\ln\left|t + \sqrt{t^2 - 4}\right| + C$$

$$= \frac{1}{2}(x+1)\sqrt{(x+1)^2 - 4}$$

$$-2\ln\left|x + 1 + \sqrt{(x+1)^2 - 4}\right| + C$$

FORM – 2

$$\int \frac{px+q}{ax^2+bx+c}\,dx$$

$$\int \frac{px+q}{\sqrt{ax^2+bx+c}}\,dx$$

$$\int (px+q)\sqrt{ax^2+bx+c}\,.\,dx$$

In problems like these, where both 1st degree and 2nd degree polynomials are present → Take the following step:

$$\boxed{(px+q) = A\frac{d}{dx}\left(ax^2+bx+c\right) + B}$$

The value of A and B can be found by comparing coefficients on both sides.

EXAMPLE →

$$\int \frac{2x+1}{x^2+2x+2}\,dx$$

$$2x+1 = A(2x+2)+B$$

$$2x+1 = 2Ax+2A+B$$

$$2A = 2 \Rightarrow A = 1$$

$$2A+B = 1 \Rightarrow B = -1$$

$$I = \int \frac{A(2x+2)+B}{x^2+2x+2}\,dx$$

$$I = \int \frac{(2x+2)-1}{x^2+2x+2}\,dx$$

$$I = \int \frac{2x+2}{x^2+2x+2}\,dx - \int \frac{1}{x^2+2x+2}\,dx$$

$$x^2 + 2x + 2 = t$$
$$(2x + 2)dx = dt$$

$$\int \frac{2x+2}{x^2+2x+2}\,dx = \int \frac{dt}{t}$$
$$= \ln|t| + C_1 = \ln|x^2 + 2x + 2| + C_1$$

$$\int \frac{1}{x^2+2x+2}\,dx =$$
$$\int \frac{1}{(x+1)^2+1}\,dx = \tan^{-1}(x+1) + C_2$$

$$I = \ln|x^2 + 2x + 2|$$
$$- \tan^{-1}(x+1) + C$$

FORM – 3

$$\boxed{\int \frac{x^2 \pm 1}{x^4 + kx^2 + 1}\, dx}$$

NOTE →

The coefficient of x^4 and the constant term must be the same

Method to Solve →

Divide both the numerator and the denominator by x^2.

EXAMPLE →

$$I = \int \frac{1 + x^2}{x^4 + 3x^2 + 1}\, dx$$

$$I = \int \frac{\frac{1}{x^2} + 1}{x^2 + 3 + \frac{1}{x^2}}\, dx$$

$$I = \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 5}\, dx$$

$$\text{Let } x - \frac{1}{x} = t$$

$$\left(1 + \frac{1}{x^2}\right) dx = dt$$

$$I = \int \frac{dt}{t^2 + (\sqrt{5})^2}$$

$$I = \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{t}{\sqrt{5}}\right) + C$$

$$I = \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{x - \frac{1}{x}}{\sqrt{5}}\right) + C$$

FORM – 4

$$\int \frac{dx}{(ax+b)\sqrt{px+q}}$$

$$\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$$

Formula to follow to solve this →

$$\boxed{px+q=t^2}$$

EXAMPLE →

$$\text{Evaluate } \int \frac{1}{(x+1)\sqrt{x+2}}\,dx$$

$$\text{Let } x+2=t^2$$

$$dx = 2t\,dt$$

$$I = \int \frac{2t\,dt}{(t^2-1)\,t}$$

$$I = 2\int \frac{1}{t^2 - 1}\,dt$$

$$I = 2\left(\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right|\right) + C$$

$$I = \ln\left|\frac{\sqrt{x+2}-1}{\sqrt{x+2}+1}\right| + C$$

FORM – 5

$$\int \frac{dx}{(px+q)\sqrt{ax^2+bx+c}}$$

$$\Rightarrow \textbf{Let px } + \textbf{ q} = \frac{\mathbf{1}}{\mathbf{t}}$$

$$\int \frac{dx}{(px^2+q)\sqrt{ax^2+b}}$$

$$\Rightarrow \textbf{Let x } = \frac{\mathbf{1}}{\mathbf{t}}$$

EXAMPLE →

$$\int \frac{1}{(x+1)\sqrt{x^2+x+1}}\,dx$$

$$\text{Let } x+1 = \frac{1}{t}$$

$$dx = -\frac{1}{t^2}\,dt$$

$$x^2+x+1 = \left(\frac{1}{t}-1\right)^2 + \frac{1}{t} - 1 + 1$$

$$x^2+x+1 = \frac{1}{t^2} - \frac{2}{t} + 1 + \frac{1}{t}$$

$$x^2+x+1=\frac{1}{t^2}-\frac{1}{t}+1$$

$$\int\frac{1}{(x+1)\sqrt{x^2+x+1}}\,dx$$

$$I=\int\frac{-1}{\frac{1}{t}\sqrt{\frac{1}{t^2}-\frac{1}{t}+1}}\left(-\frac{1}{t^2}\,dt\right)$$

$$I=\int\frac{-1}{\frac{1}{t}\sqrt{\frac{1-t+t^2}{t^2}}}\left(-\frac{1}{t^2}\,dt\right)$$

$$I=\int\frac{-1}{\frac{1}{t^2}\sqrt{t^2-t+1}}\left(-\frac{1}{t^2}\,dt\right)$$

$$I=\int\frac{dt}{\sqrt{t^2-t+1}}$$

$$t^2 - t + 1 = \left(t - \tfrac{1}{2}\right)^2 + \tfrac{3}{4}$$

$$I = \int \frac{dt}{\sqrt{t^2 - t + 1}}$$

$$I = \int \frac{dt}{\sqrt{\left(t - \tfrac{1}{2}\right)^2 + \tfrac{3}{4}}}$$

$$I = \int \frac{dt}{\sqrt{\left(t - \tfrac{1}{2}\right)^2 + \left(\tfrac{\sqrt{3}}{2}\right)^2}}$$

$$\int \frac{dx}{\sqrt{a^2 + x^2}}$$

$$= \log_e \left| x + \sqrt{a^2 + x^2} \right| + c$$

$$I = \ln \left| t - \tfrac{1}{2} + \sqrt{t^2 - t + 1} \right| + C$$

$$I = \ln\left|\frac{1}{x+1} - \frac{1}{2} + \sqrt{\left(\frac{1}{x+1}\right)^2 - \frac{1}{x+1} + 1}\right| + C$$

FORM – 6

$$\boxed{I = \int \frac{dx}{x(x^n + 1)}}$$

In such integrals, take the highest power of x in the denominator common.

$$I = \int \frac{dx}{x.\,x^{\,n}\left(1 + x^{-n}\right)}$$

$$I = \int \frac{dx}{x^{\,n+1}\left(1 + x^{-n}\right)}$$

$$1 + x^{-n} = 1 + \frac{1}{x^n} = t$$

$$-n\frac{1}{x^{n+1}}dx = dt \to \frac{dx}{x^{n+1}} = -\frac{dt}{n}$$

$$I = \int \frac{-dt}{n.t} = \frac{-1}{n} \int \frac{1}{t} dt$$

$$= \frac{-1}{n} \log_e |t| + c$$

$$= \frac{-1}{n} \log_e \left| 1 + \frac{1}{x^n} \right| + c$$

EXAMPLE →

$$\text{Evaluate } \int \frac{\cos x \, dx}{\sin^8 x + \sin x}$$

$$\text{Let } \sin x = t$$

$$\cos x \, dx = dt$$

$$\int \frac{\cos x \, dx}{\sin^8 x + \sin x} = \int \frac{dt}{t^8 + t}$$

$$I = \int \frac{dt}{t^8\left(1 + \frac{1}{t^7}\right)}$$

$$I = \int \frac{dt}{t^8\left(1 + \frac{1}{t^7}\right)}$$

$$1 + \frac{1}{t^7} = z \rightarrow \frac{-7}{t^8}dt = dz$$

$$I = \int \frac{dz}{-7\,(z)} = -\frac{1}{7}\ln|z| + C$$

$$I = -\frac{1}{7}\ln\left|1 + \frac{1}{\sin^7 x}\right| + C$$

Integration of Trigonometric Functions

FORM – 7

$$\int \frac{dx}{a + b\sin^2 x}$$

$$\int \frac{dx}{a + b\cos^2 x}$$

$$\int \frac{dx}{a\cos^2 x + b\sin x\cos x + c\sin^2 x}$$

Divide the numerator and denominator by $\cos^2 x$ and substitute $\tan x = t$

EXAMPLE →

$$\int \frac{dx}{1 + 2\sin^2 x}$$

$$= \int \frac{\frac{1}{\cos^2 x}\,dx}{\sec^2 x + 2\tan^2 x}$$

$$= \int \frac{\sec^2 x\,dx}{\sec^2 x + 2\tan^2 x}$$

$$= \int \frac{\sec^2 x\,dx}{1 + \tan^2 x + 2\tan^2 x}$$

$$= \int \frac{\sec^2 x\,dx}{1 + 3\tan^2 x}$$

$$= \int \frac{\sec^2 x\,dx}{\sec^2 x + 2\tan^2 x}$$

$$(\text{Let } \tan x = t \Rightarrow dt = \sec^2 x\,dx)$$

$$= \int \frac{dt}{1 + 3t^2}$$

$$\left(\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c\right)$$

$$= \frac{1}{3}\int \frac{dt}{\left(\frac{1}{\sqrt{3}}\right)^2 + t^2}$$

$$= \frac{1}{3}\cdot\frac{1}{\frac{1}{\sqrt{3}}}\tan^{-1}\left(\frac{t}{1/\sqrt{3}}\right) + C$$

$$= \frac{1}{\sqrt{3}}\tan^{-1}(\sqrt{3}\tan x) + C$$

FORM – 8

$$\int \frac{dx}{a + b\sin x}$$

$$\int \frac{dx}{a + b\cos x}$$

$$\int \frac{dx}{a\sin x + b\cos x + c}$$

(i) Convert $\sin x$ and $\cos x$

in terms of $\tan\left(\frac{x}{2}\right)$

(ii) Substitute $\tan\left(\frac{x}{2}\right) = t$

$$\sin x = \frac{2\tan\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)}$$

$$\cos x = \frac{1 - \tan^2\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)}$$

$$\text{Evaluate } \int \frac{dx}{1+\sin x+\cos x}$$

$$I = \int \frac{dx}{\frac{1+2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} + \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$$

$$I = \int \frac{1+\tan^2\frac{x}{2}}{2+2\tan\frac{x}{2}}\,dx$$

$$I = \int \frac{1+\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}+2\tan\frac{x}{2}+1-\tan^2\frac{x}{2}}\,dx$$

$$\text{Let } 2+2\tan\frac{x}{2} = t$$

$$\rightarrow \frac{2}{2}\sec^2\frac{x}{2}\,dx = dt$$

$$\sec^2\frac{x}{2}\,dx = dt$$

$$I = \int \frac{\sec^2 \frac{x}{2}\, dx}{2 + 2\tan \frac{x}{2}} = \int \frac{dt}{t}$$

$$I = \ln|t| + C = \ln\left|2 + 2\tan \frac{x}{2}\right| + C$$

FORM – 9

$$\int \frac{p\cos x + q\sin x + r}{a\cos x + b\sin x + c}\, dx$$

$$\boxed{\begin{gathered}\text{Write Numerator} = A(\text{Denominator}) \\ +B\left(\frac{d}{dx}(\text{Denominator})\right) + C\end{gathered}}$$

Compare coefficients on both sides.

$$\text{Evaluate: } \int \left(\frac{5\tan x\, dx}{\tan x - 2}\right)$$

$$\int \frac{5\tan x\,dx}{\tan x - 2} = \int \frac{5\sin x}{\sin x - 2\cos x}\,dx$$

$$\frac{5\sin x}{\sin x - 2\cos x} = \frac{1(\sin x - 2\cos x)}{(\sin x - 2\cos x)} + \frac{2(\cos x + 2\sin x)}{(\sin x - 2\cos x)}$$

$$\frac{5\sin x}{\sin x - 2\cos x} = 1 + \frac{2(\cos x + 2\sin x)}{(\sin x - 2\cos x)}$$

$$5\sin x = A(\sin x - 2\cos x) + B(\cos x + 2\sin x) + C$$

$$-2A + B = 0 \rightarrow A = 1$$

$$A + 2B = 5 \rightarrow B = 2$$

$$C = 0$$

$$I = \int \frac{5\sin x}{\sin x - 2\cos x}\,dx = \int 1.dx$$

$$+ \int \frac{2(\cos x + 2\sin x)}{(\sin x - 2\cos x)}dx$$

$$I = \int 1\,dx + 2\int \frac{\cos x + 2\sin x}{\sin x - 2\cos x}\,dx$$

$$I = x + 2\int \frac{dt}{t}$$

$$I = x + 2\ln|t| + K$$

$$I = x + 2\ln\left|\sin x - 2\cos x\right| + K$$

FORM – 10

$$\int \frac{\cos x \pm \sin x}{f(\sin 2x)}\, dx$$

$$\text{Put } \int \text{Numerator} = t$$

$$\text{Evaluate: } I = \int \frac{\cos\theta - \sin\theta}{\sqrt{8 - \sin 2\theta}}\, d\theta$$

$$\text{Let } t = \sin\theta + \cos\theta$$

$$\Rightarrow\ dt = (\cos\theta - \sin\theta)\, d\theta$$

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta$$
$$+2\sin\theta\cos\theta = 1 + \sin 2\theta$$

$$\Rightarrow\ \sin 2\theta = t^2 - 1$$

Hence $\sqrt{8-\sin 2\theta} = \sqrt{8-(t^2-1)}$

$$= \sqrt{9-t^2} = \sqrt{3^2-t^2}$$

The integral becomes $I = \displaystyle\int \frac{dt}{\sqrt{9-t^2}}$

$$\left(\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C\right)$$

$$\Rightarrow\ I = \sin^{-1}\left(\frac{t}{3}\right) + C$$

$$I = \sin^{-1}\left(\frac{\sin\theta + \cos\theta}{3}\right) + C$$

FORM – 11

$$\int \sin^m x \ \cos^n x \, dx$$

CASE 1 – Both the Powers are Even

To solve such equations, it is important to remove the power somehow.

$$I = \int \sin^2 x \cos^4 x \, dx$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\cos^4 x = (\cos^2 x)^2 = \left(\frac{1 + \cos 2x}{2}\right)^2$$

$$I = \int \sin^2 x \cos^4 x \, dx =$$

$$\int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right)^2 dx$$

$$I = \frac{1}{8} \int (1 - \cos 2x)(1 + \cos 2x)^2 \, dx$$

$$(1 - \cos 2x)(1 + \cos 2x)^2 =$$

$$(1 - \cos 2x)(1 + 2\cos 2x + \cos^2 2x)$$

$$= 1 + 2\cos 2x + \cos^2 2x$$

$$- \cos 2x - 2\cos^2 2x - \cos^3 2x$$

$$= 1 + \cos 2x - \cos^2 2x - \cos^3 2x$$

$$\cos^2 2x = \frac{1 + \cos 4x}{2}$$

$$\cos^3 2x = \cos 2x \cdot \cos^2 2x$$

$$= \cos 2x \cdot \frac{1+\cos 4x}{2}$$

$$I = \frac{1}{8}\int (1-\cos 2x)(1+\cos 2x)^2\,dx$$

$$I = \frac{1}{16}\int \left[1+\cos 2x - \frac{1+\cos 4x}{2} - \cos 2x \cdot \frac{1+\cos 4x}{2}\right] dx$$

$$I = \frac{1}{16}\left[x + \frac{\sin 2x}{2} - \frac{\sin 4x}{8} - \frac{1}{2}\left(\frac{\sin 2x}{1} + \frac{\sin 6x}{6}\right)\right] + C$$

$$I = \frac{1}{16}\left[x + \frac{\sin 2x}{2} - \frac{\sin 4x}{8} - \frac{\sin 2x}{2} - \frac{\sin 6x}{12}\right] + C$$

CASE 2 – At least one Power is Odd

$$\int \sin^m x\ \cos^n x\,dx$$

Method to Follow →

- **If m = odd → put cos x = t**
- **If n = odd → put sin x = t**
- **If both are odd → put sin x = cos x = t**

$$I = \int \sin^3 x \cos^2 x \, dx$$

$$Let \cos x = t \Rightarrow - \sin x \, dx = dt$$

$$I = \int (\sin x) \ \sin^2 x \ \cos^2 x \, dx$$

$$= - \int (1 - t^2) \, t^2 \, dt$$

$$= - \int (t^2 - t^4) \, dt$$

$$= - \left[\frac{t^3}{3} - \frac{t^5}{5} \right] + C$$

$$= - \left[\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} \right] + C$$

CASE 3 → m + n is a negative even integer

$$\int \sin^m x \ \cos^n x \, dx$$

Method to Follow:

put → tan x = t

$$\text{Evaluate } \int \sec^{\frac{2}{3}} x \ \csc^{\frac{4}{3}} x \, dx$$

$$\int \cos^{-\frac{2}{3}} x \, \sin^{-\frac{4}{3}} x \, dx$$

$$m + n = -\frac{2}{3} - \frac{4}{3} = -2 \Rightarrow \tan x = t$$

$$t = \tan x \rightarrow dt = \sec^2 x \, dx$$

$$\int \frac{1}{\cos^{\frac{2}{3}} x} \ \frac{1}{\sin^{\frac{4}{3}} x} \, dx$$

$$= \int \frac{1}{\cos^{\frac{6}{3}} x} \frac{1}{\frac{\sin^{\frac{4}{3}} x}{\cos^{\frac{4}{3}} x}} dx$$

$$= \int \frac{1}{\cos^2 x} \frac{1}{\tan^{\frac{4}{3}} x} dx$$

$$= \int \frac{\sec^2 x}{\tan^{\frac{4}{3}} x} dx$$

$$= \int \frac{dt}{t^{\frac{4}{3}}} = \frac{t^{\frac{-4}{3}+1}}{\frac{-4}{3}+1} + c = \frac{t^{\frac{-1}{3}}}{\frac{-1}{3}} + c$$

$$= -3t^{\frac{-1}{3}} + c = -3\left(\tan x\right)^{\frac{-1}{3}} + c$$

Integration Using Partial Fractions

FORM – 1 (Nonrepeating Linear Factors)

$$\frac{1}{(x-1)(x-2)} = \frac{a}{x-1} + \frac{b}{x-2}$$

$$\frac{x-1}{(x-2)(x-3)} = \frac{a}{(x-2)} + \frac{b}{(x-3)}$$

$$\frac{x-2}{(x-1)(x-3)(x-4)} = \frac{a}{x-1} + \frac{b}{x-3} + \frac{c}{x-4}$$

FORM – 2 (Repeating Linear Factors)

$$\frac{1}{(x-1)(x-2)^2} = \frac{a}{x-1} + \frac{b}{x-2} + \frac{c}{(x-2)^2}$$

FORM – 3 (One Unfactorizable Factor)

$$\frac{1}{(x-1)(x^2+2)} = \frac{a}{x-1} + \frac{bx+c}{x^2+2}$$

FORM – 4 (Repeating Unfactorizable Factor)

$$\frac{1}{(x^2+2)^2(x-1)} = \frac{a}{x-1} + \frac{bx+c}{x^2+2} + \frac{dx+e}{(x^2+2)^2}$$

$$\int \frac{x-1}{(x-2)(x-3)}dx = \int \left(\frac{-1}{x-2} + \frac{2}{x-3}\right)dx$$

$$\frac{x-1}{(x-2)(x-3)} = \frac{a}{x-2} + \frac{b}{x-3}$$

$$x - 1 = ax - 3a + bx - 2b$$

$$-3a - 2b = -1$$

$$a + b = 1$$

$$a = -1\,,\; b = 2$$

$$\int \frac{1}{(x-1)(x-2)}dx = \int \left(\frac{-1}{x-1} + \frac{1}{x-2}\right)dx$$

$$= -\int \frac{1}{x-1}dx + \int \frac{1}{x-2}dx$$

$$= \ln|x-1| + \ln|x-2| + C$$

$$\frac{1}{(x-1)(x-2)} = \frac{a}{(x-1)} + \frac{b}{(x-2)}$$

$$0x + 1 = ax - 2a + bx - b$$

$$a + b = 0$$

$$-2a - b = 1$$

$$a = -1\,,\; b = 1$$

$$\text{Evaluate} \int \frac{2x-1}{(x+1)(x^2+2)}dx$$

$$= \int \frac{-1}{x+1}dx + \int \frac{x+1}{x^2+2}dx$$

$$= -\ln|x+1| + \frac{1}{2}\int \frac{2x}{x^2+2}dx + \int \frac{1}{x^2+2}dx$$

$$= -\ln|x+1| + \frac{1}{2}\ln|x^2+2| + \frac{1}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}} + C$$

$$\frac{2x-1}{(x+1)(x^2+2)} = \frac{a}{x+1} + \frac{bx+c}{x^2+2}$$

$$2x-1 = ax^2 + 2a + bx^2 + bx + cx + c$$

$$a+b=0 \quad b+c=2 \quad 2a+c=-1$$

$$a=-1\,,\, b=1\,,\, c=1$$

Reduction Formulas for Trigonometry

$$I_n = \int \sin^n x\, dx, \quad n \geq 2$$
$$= -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$$

$$I_n = \int \cos^n x\, dx, \quad n \geq 2$$
$$= \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$$

$$I_n = \int \tan^n x\, dx, \quad n \geq 2$$
$$\implies I_n + I_{n-2} = \frac{\tan^{n-1} x}{n-1}$$

$$I_n = \int \cot^n x \, dx, \quad n \geq 2$$

$$\implies I_n + I_{n-2} = -\frac{\cot^{n-1} x}{n-1}$$

$$I_n = \int \sec^n x \, dx, \quad n \geq 2$$

$$\implies I_n = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

$$I_n = \int \csc^n x \, dx, \quad n \geq 2$$

$$\implies I_n = -\frac{\cot x \csc^{n-2} x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

Non-Trigonometric Reduction Formulas

$$I_n = \int x^n e^x \, dx = x^n e^x - n I_{n-1}$$

$$1.\ I_n = \int \sin^n x\,dx = \int (\sin x)^{n-1}\ \sin x\,dx$$

$$u = (\sin x)^{n-1} \qquad v = \sin x$$

$$2.\ I_n = \int \cos^n x\,dx = \int (\cos x)^{n-1}\ \cos x\,dx$$

$$u = (\cos x)^{n-1} \qquad v = \cos x$$

$$3.\ I_n = \int \tan^n x\,dx = \int (\tan x)^{n-2}\ \tan^2 x\,dx$$

$$\text{Let } \tan x = t \qquad \sec^2 x\,dx = dt$$

$$4.\ I_n = \int \cot^n x\,dx = \int (\cot x)^{n-2}\ \cot^2 x\,dx$$

$$\text{Let } \cot x = t \qquad -\ \csc^2 x\,dx = dt$$

$$5.\ I_n = \int \sec^n x\,dx = \int (\sec x)^{n-2}\ \sec^2 x\,dx$$

$$u = (\sec x)^{n-2} \qquad v = \sec^2 x$$

$$\text{Formulas used later: } \tan^2 x = \sec^2 x - 1$$

$$\int \sec x\,dx = \log_e |\sec x + \tan x|$$

$$6.\ I_n = \int \csc^n x\,dx = \int (\csc x)^{n-2}\ \csc^2 x\,dx$$

$$u = (\csc x)^{n-2} \qquad v = \csc^2 x$$

$$\text{Formulas used later: } \cot^2 x = \csc^2 x - 1$$

$$\int \csc x\,dx = \log_e \left|\tan \frac{x}{2}\right|$$